Statistical and Computational Models

Lecture 12: Duration and event history

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Roadmap

Durations

Regression for durations

Durations as outcomes

Very often we encounter **durations**, the time until an event:

$$T_i \in [0, \infty)$$

Examples in political science:

- Civil wars, until they end
- Cabinets, until they dissolve
- Regimes, until they collapse

The survival function

The **survival function** – $S(t)$ – is the chance a unit lasts past time t.

Recall the CDF, $F(t) = \mathbb{P}(T_i \leq t)$:

$$S(t) = \mathbb{P}(\underbrace{T_i > t}_{\text{still going at } t}) = 1 - \mathbb{P}(\underbrace{T_i \leq t}_{\text{ended by } t})$$

For example, $S(20)$ evaluates the chance a regime is still in place 20 years after it began.

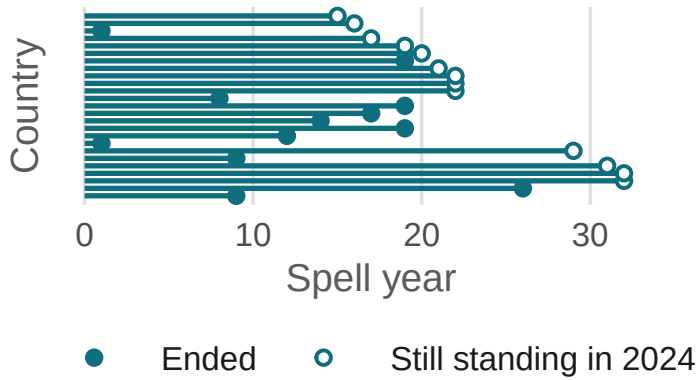
Note: $S(0) = 1$, and $S(t)$ never rises.

We want to estimate $S(t)$ from data, and how it differs between units, based on their covariates.

Censoring

Suppose we also recorded each regime's start, how long it lasted or had lasted by 2024 (`time`), and whether it ended (`status`):

country	start_year	time	status	lgdp	oil
15	2005	20	0	9.23	0
16	1998	12	1	8.32	0
17	2003	8	1	6.49	1
18	2000	17	1	7.56	1



Put each regime on its own clock: each line is a **spell**, from a regime's start to its end, or the final year of observation.

The open ones are **right-censored**: we know only that they lasted at least this long, not when they will end.

Assume **independent censoring**: given its covariates, when we stop watching a unit says nothing about when its event would come.

Kaplan–Meier

To get at $S(t)$, go one spell year at a time, and just count:

j	Standing r_j	Ended q_j	$\hat{h}_j$
1	205	7	0.034
2	198	2	0.010
3	196	3	0.015

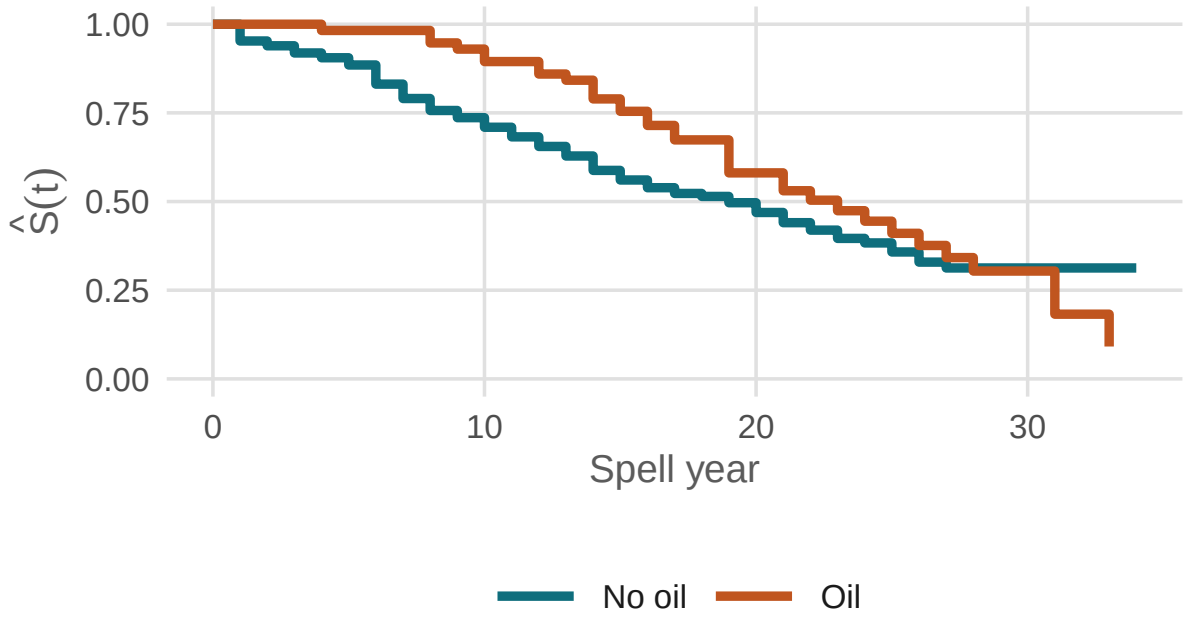
The **yearly hazard** h_j is the chance of ending in spell year j, given still standing when it began; $\hat{h}_j = q_j / r_j$.
A regime still standing in 2024 counts in r_j every year we saw it, and in no q_j.

To last past year 20, a regime must get through every spell year up to it, each given the ones before:

$$S(20) = \underbrace{\mathbb{P}(T_i > 1)}_{1-h_1} \underbrace{\mathbb{P}(T_i > 2 \mid T_i > 1)}_{1-h_2} \cdots \underbrace{\mathbb{P}(T_i > 20 \mid T_i > 19)}_{1-h_{20}}$$

Plug in each $\hat{h}_j$: the **Kaplan–Meier** estimator, $\hat{S}(t) = \prod_{j \le t}(1 - \hat{h}_j)$. Here $\hat{S}(20) = 0.500$.

Kaplan–Meier with covariates



For a discrete covariate, we can compute $\hat{S}(t)$ separately at each of its levels: in our case, with and without `oil`.

Kaplan–Meier is **non-parametric**: it assumes no shape for $S(t)$, it only counts. Like local averaging, it needs enough units at every level: for `lgdp`, or several covariates at once, the groups get too thin.

Roadmap

Durations

Regression for durations

Durations as binary outcomes

For a larger number of Xs, or continuous Xs, we need a parametric model – regression. First, reshape the data to country-spell year level.

country	time (j)	lgdp	oil	status
15	1	9.23	0	0
15	2	9.23	0	0
⋮	⋮	⋮	⋮	⋮
15	20	9.23	0	0
16	1	8.32	0	0
⋮	⋮	⋮	⋮	⋮
16	12	8.32	0	1

```
library(survival)
country_years <- survSplit(
  Surv(time, status) ~ .,
  data = countries,
  cut = 1:max(countries$time),
  id = "country")
nrow(country_years)
```

```
[1] 3492
```

The outcome is `status`:

$$y_{ij} = \begin{cases} 1 & \text{regime } i \text{ ended in spell year } j \\ 0 & \text{it got through spell year } j \end{cases}$$

for each spell year we saw

A discrete-time hazard model

Recall the **two choices** for a GLM: the outcome's distribution, and a link that connects the linear model $\boldsymbol{x}_i^\top \beta$ to its parameter.

Assume a Bernoulli and the **complementary log-log**:

$$y_{ij} \sim \text{Bernoulli}(h_{ij})$$
$$\log\Big(-\log(1-h_{ij})\Big) = \gamma_j + \boldsymbol{x}_i^\top \beta$$

where h_{ij} is the yearly hazard: the chance regime i ends in spell year j, given it was still standing when the year began.

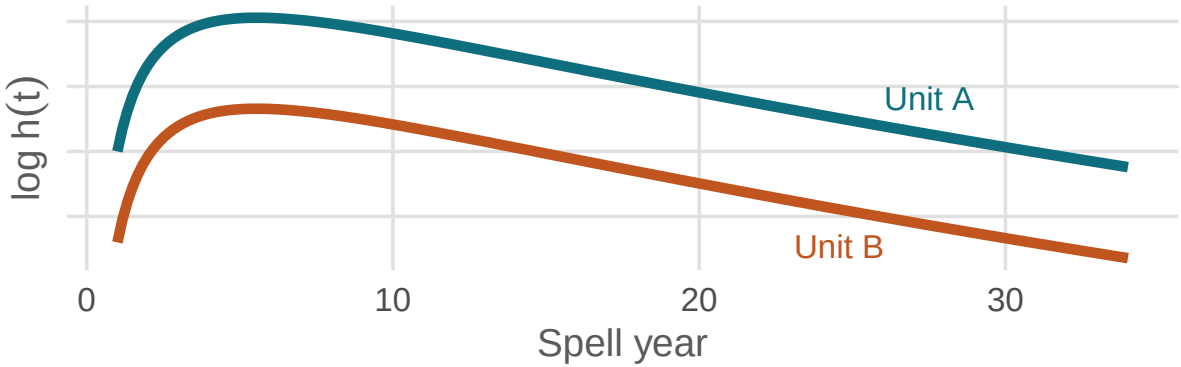
One intercept γ_j per spell year lets the baseline take any shape: a **discrete-time hazard model**.

If we have no covariates, only the γ_j, what do we get? Kaplan–Meier's $\hat{h}_j = q_j / r_j$.

Proportional hazards

Note that β is not indexed by j: a one-unit increase in x_{im} multiplies the hazard by the same factor, e^{β_m}, in every spell year.

This is the **proportional hazards** assumption, which forces any two units' log hazards to run parallel:



Aside: why the cloglog, not the logit? Think of regime-ending shocks in spell year j as a Poisson count with rate $\lambda_{ij} = \exp(\gamma_j + x_i^\top \beta)$, the Poisson's log link. A regime gets through the year if no shock comes: $1 - h_{ij} = e^{-\lambda_{ij}}$. Take logs twice, and $\log\left(-\log(1 - h_{ij})\right) = \gamma_j + x_i^\top \beta$: the cloglog. So e^{β_m} multiplies the rate λ_{ij}; when h_{ij} is small, $\lambda_{ij} \approx h_{ij}$.

Fitting a discrete-time hazard model in R

Using `glm()` on the country-years:

```
fit_cloglog <- glm(status ~ factor(time) + lgdp + oil,
                   family = binomial(link = "cloglog"),
                   data = country_years)
round(coef(summary(fit_cloglog))[c("lgdp", "oil"), 1:2], 2)
```

```
     Estimate Std. Error
lgdp    -0.73       0.09
oil     -0.53       0.21
```

Note: `factor(time)` gives the γ_j (not printed).

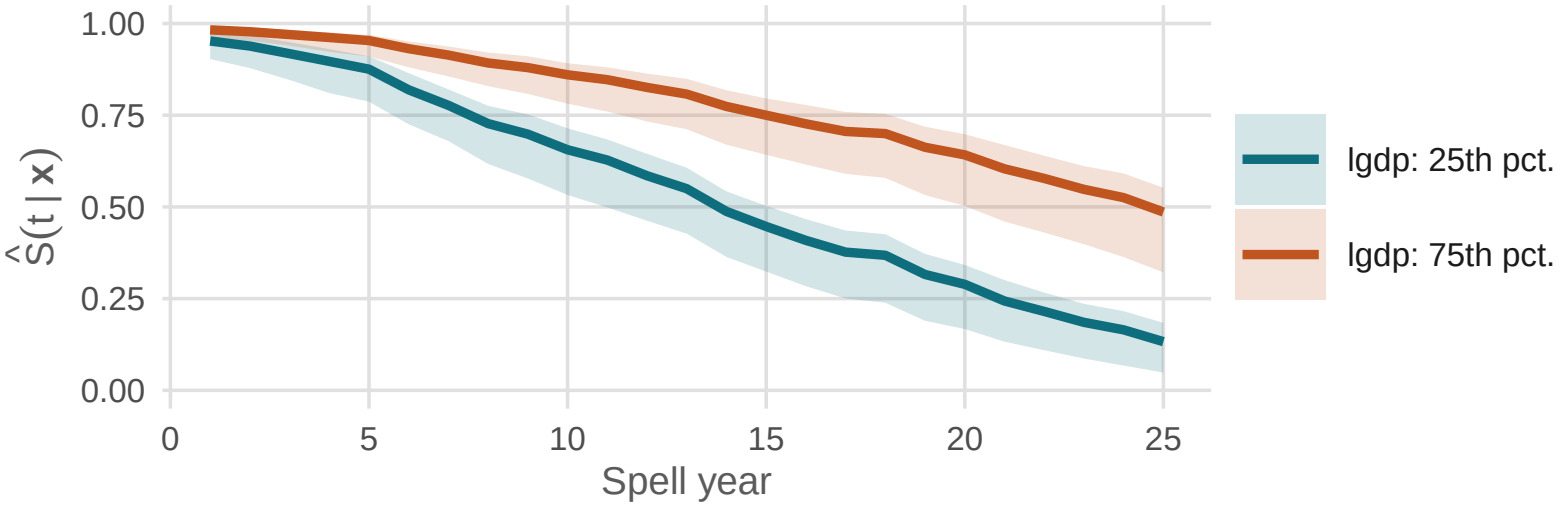
The rows are the **coefficients** $\hat{\beta}$, which, given the cloglog link, linearly map $\boldsymbol{x}_i$ to the log of the hazard rate.

1. Read the **sign**: negative lowers the hazard, so the spell runs longer.
2. Exponentiate for a **hazard ratio**: $e^{\hat{\beta}_{\text{lgdp}}} = 0.48$ per unit of `lgdp`.
3. For a change of size c, $e^{c\hat{\beta}_m}$: doubling GDP per capita ($c = \log 2$) gives 0.60.

Interpreting `lgdp`, *ceteris paribus*

As with every GLM, we want **quantities of interest** in the outcome's units: here, the chance a regime is still standing.

Chain a country's predicted yearly hazards, as Kaplan–Meier did: $\hat{S}(t \mid \boldsymbol{x}) = \prod_{j \le t} \left(1 - \hat{h}_j(\boldsymbol{x})\right)$. Two countries without oil, `lgdp` at its 25th and 75th percentiles:



Band: 95% simulation interval. In `R`, `predictions()` on a country's rows, then `cumprod(1 - estimate)`.

A more flexible parametric model

Proportional hazards forces one ratio for `oil` onto every spell year. We can relax this by allowing it to change, through an **interaction** with $\log j$:

$$\log\Big(-\log(1-h_{ij})\Big) = \gamma_j + \beta_1\,\text{oil}_i + \beta_2\,\text{oil}_i \log j + \beta_3\,\text{lgdp}_i$$

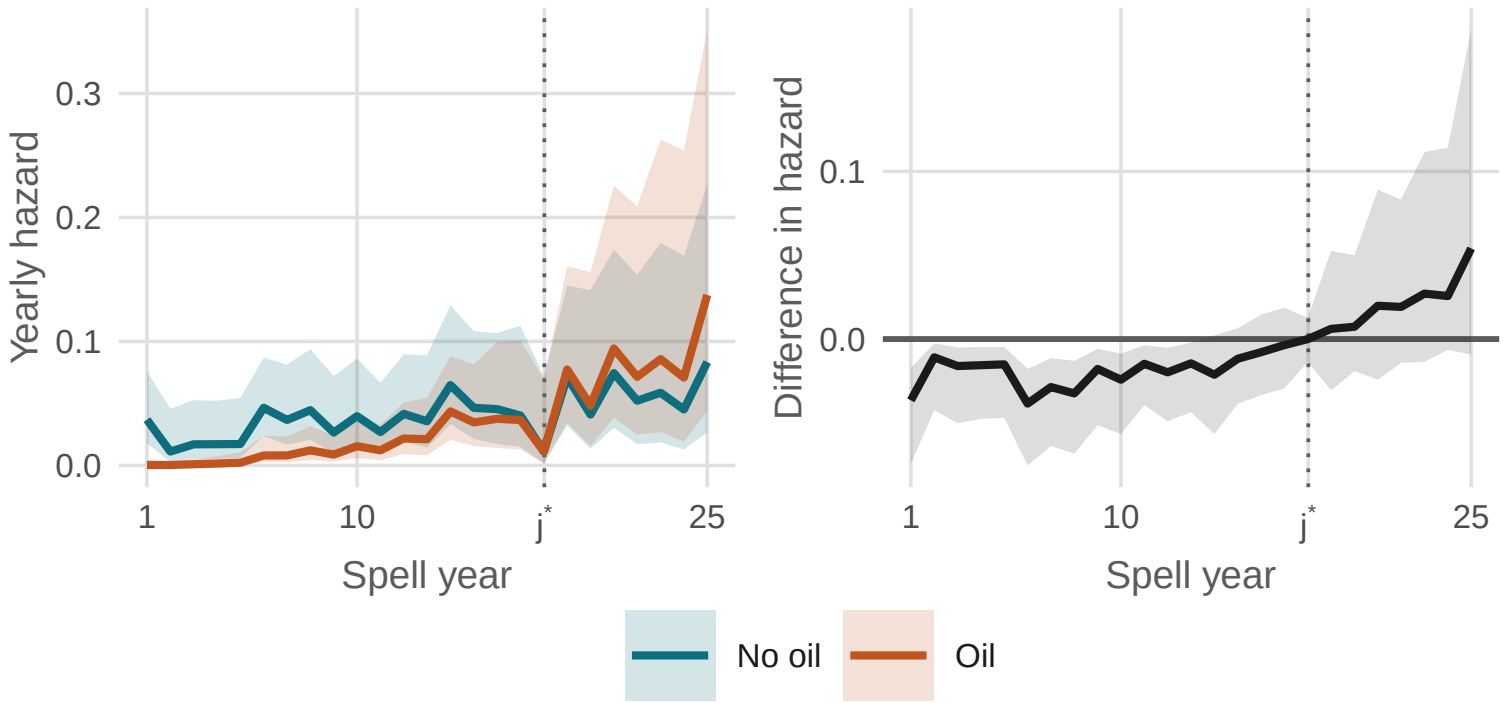
In `R`, add `oil:log(time)` to the `glm()` formula.

Oil's hazard ratio in spell year j is now $\exp(\beta_1 + \beta_2 \log j)$. It equals 1 at

$$j^* = \exp(-\beta_1/\beta_2)$$

Interpreting oil, *ceteris paribus*

A country at median `lgdp`, without and with oil, from the time-varying model:



Left: yearly hazards. Right: their difference, oil minus no oil. Bands: 95% simulation intervals.

Fitting GLMs in R

Family or function	Link	Variance	Model
`gaussian`	identity	σ^2	normal linear
`binomial`	logit (or probit)	$\mu(1-\mu)$	logit, probit
`poisson`	log	μ	Poisson
`quasipoisson`	log	$\phi\mu$	overdispersed Poisson
`MASS::glm.nb`	log	$\mu + \mu^2/\theta$	negative binomial
`MASS::polr`	cumulative logit	–	ordered logit
`nnet::multinom`	baseline-category logit	–	multinomial logit
`binomial`	cloglog	$\mu(1-\mu)$	discrete-time hazard
`survival::coxph`	log, of the hazard	–	Cox

`{marginaleffects}` and `{modelsummary}` take both. `predictions()` gives yearly hazards $\hat{h}_{ij}$ on the cloglog, and survival curves on `coxph()` (`type = "survival"`).

In `modelsummary()`: `exponentiate = TRUE` for hazard ratios, `coef_omit = "time"` to hide the $\hat{\gamma}_j$.

Other duration models

- **Parametric baselines**: restrict the γ_j. $\gamma_j = \gamma$ is a flat hazard (exponential); $\gamma_j \approx \gamma + (\alpha - 1) \log j$ only rises or only falls (Weibull). In our `glm()`, drop `factor(time)`, or swap it for `log(time)`.
- **Cox model**: estimates β without the baseline. It keeps only which unit had each event, among those still standing, not when. Its β are the same log hazard ratios as ours.
- **Competing risks**: suppose a regime can end in a coup or an election, we fit a model that has one hazard for each possible event.
- **Random survival forests**: tree-based method, where each leaf estimates a survival curve, not a single class.

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **11.1–11.4**, 11.5. statlearning.com.
- Licht, A. A. (2011) "Change Comes with Time: Substantive Interpretation of Nonproportional Hazards in Event History Analysis." *Political Analysis* 19(2): 227–243.
- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – ch. 7. grodri.github.io/glms.